「The Primary decomposition theorem.
Let T be a linear operator on the finite-dimensional vector space V over a field F.
Let P be the minimal polynomial for T with the irreducible decomposition

$$P = P_1^{r_1} \cdot P_2^{r_2} \cdots P_k^{r_k}$$

Then,

$$V = \bigoplus_{i=1}^{k} \ker P_i^{r_i}(T) \text{ and each } \ker P_i^{r_i}(T) \text{ is invariant } T.$$

Moreover, the restriction of T by $\ker P_i^{r_i}(T)$ has the minimal polynomial as $P_i^{r_i}$.
Proof. For each $1 \le i \le k$, define

$$f_i = \prod_{j \ne i} P_j^{r_j}$$

Since P_i's are prime, $f_1, \cdots, f_k$ are relatively prime,

$$f_1 g_1 + \cdots + f_k g_k = 1 \text{ for some polynomials } g_1, \cdots, g_k.$$

And, clearly, for $i \ne j$, $f_i f_j$ is divided by P.
For each $1 \le i \le k$, Put $E_i := f_i(T) g_i(T)$. Then,

$$E_1 + E_2 + \cdots + E_k = I \text{ and } E_i E_j = 0 \text{ if } i \ne j,$$

because $f_i f_j$ is divided by P, so is $f_i f_j g_i g_j$. Thus each E_i is a projection.」
In fact, these $E_1, \cdots, E_k$ are resolution of identity, these images decompose V.
And, to show $\operatorname{Im} E_i = \ker P_i^{r_i}$ for each $1 \le i \le k$, observe that

$$\alpha \in \operatorname{Im} E_i \Rightarrow E_i(\alpha) = \alpha \Rightarrow P_i(T)^{r_i}\alpha = P_i(T)^{r_i} E_i(\alpha) = P_i(T)^{r_i} \cdot f_i(T) \cdot g_i(T)\alpha = 0.$$

So $\alpha \in \ker P_i^{r_i}(T)$. Conversely,

$$\alpha \in \ker P_i^{r_i}(T) \Rightarrow f_j(T) g_j(T)(\alpha) = 0 \text{ for } j \ne i \Rightarrow E_i(\alpha) = \alpha$$

So $\alpha \in \operatorname{Im} E_i$. Therefore, $\operatorname{Im} E_i = \ker P_i^{r_i}(T)$ for all $1 \le i \le k$.」
The invariantness is given by: $\alpha \in \ker P_i^{r_i}(T) \Leftrightarrow P_i^{r_i}(T)\alpha = 0 \Rightarrow P_i^{r_i}(T) T\alpha = T \cdot P_i^{r_i}(T)\alpha = 0$.」
Let $T_i : \ker P_i^{r_i}(T) \to \ker P_i^{r_i}(T) : \alpha \mapsto T\alpha$, for each $1 \le i \le k$.
Clearly, $P_i^{r_i}(T_i) = 0$, and, if $g(T) = 0$, then $g(T) f_i(T) = 0$, so $g f_i$
So $g f_i$ is divided by $P_i^{r_i} \cdot f_i$, so g is divided by $P_i^{r_i}$. ☒

Thm. Let T be a linear operator on the finite dimensional V over a field V.
If the minimal polynomial of T splits over F, then there exist diagonalizable D, nilpotent N, satisfying $T = D + N$ and $DN = ND$.

Moreover, such D and N are unique, and both are polynomial in T.

Proof. Let $P(x) = (x - c_1)^{r_1} \cdots (x - c_k)^{r_k}$ be the minimal polynomial of T.
By the Primary decomposition Theorem, there exist resolution of identity:

$$E_1 + \cdots + E_k = I \text{ and } E_i E_j = \delta_{ij} E_i \text{ satisfying}$$

Each $1 \le i \le k$, $\operatorname{Im} E_i = \ker(T - c_i I)^{r_i}$. Define $D = c_1 E_1 + \cdots + c_k E_k$ and $N = T - D$.
Then,

$$T = TE_1 + \cdots + TE_k$$

$$D = c_1 E_1 + \cdots + c_k E_k$$

$$N = (T - c_1 I)E_1 + \cdots + (T - c_k I)E_k.$$

By the orthogonality, for $r = \max\{r_1, \cdots, r_k\}$,

$$N^r = (T - c_1 I)^r E_1 + \cdots + (T - c_k I)^r E_k = 0,$$

because each $1 \le i \le k$, $(x - c_i)^{r_i}$ is the minimal polynomial of the restriction $T|_{\operatorname{Im} E_i}$.
So N is nilpotent, and D is diagonalizable, directly.